

Project 2: Malicious Email (Phishing) Investigation & Triage Analysis

Overview

This project demonstrates the complete investigation of a suspected phishing email using industry-standard security tools and email analysis techniques. The objective was to identify phishing indicators, verify email authentication mechanisms, extract indicators of compromise (IOCs), analyze network artifacts, and document the findings in a structured incident report. The investigation followed a standard SOC analyst workflow from initial email inspection to final risk assessment.

What is a Phishing Email?

A phishing email is a fraudulent message designed to trick users into revealing sensitive information, such as usernames, passwords, banking details, or other confidential data. It often appears to come from a trusted organization and uses urgency or fear to convince the recipient to take immediate action.

What Happens If You Click a Phishing Email?

Clicking a phishing link can redirect you to a fake website that steals your login credentials or personal information. In some cases, it may also download malicious files, install malware, or give attackers unauthorized access to your device and accounts. This can lead to data theft, financial loss, or compromise of an organization's network.

Objective

The objective of this project was to analyze a suspicious email, determine whether it was a phishing attempt, identify malicious indicators, and document the investigation process using security analysis tools commonly used in Security Operations Centers (SOC).

Tools Used:

1. Any.Run
2. CyberChef
3. Urlscan.io
4. VirusTotal
5. MX Toolbox email Header Analyzer
6. EML Analyzer

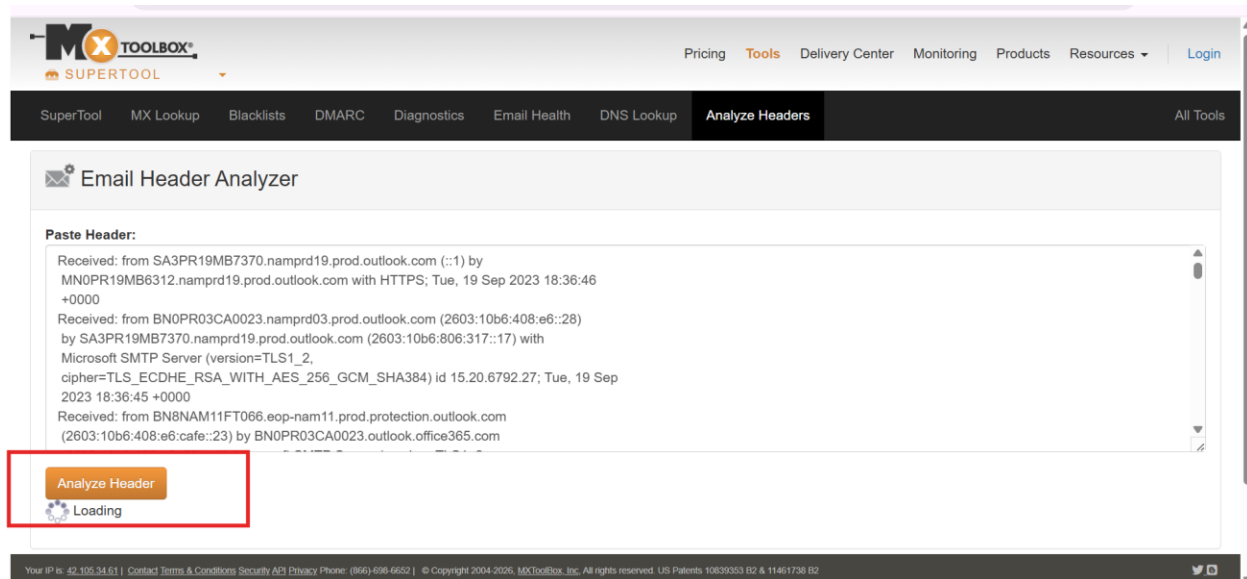
Investigation Workflow:

File For Investigation:



Step 1 – Email Collection

The investigation started by collecting the suspicious email in **.eml** format. This preserved the original email headers, metadata, and message content, allowing a complete forensic analysis without altering the evidence.



Step 2 – Email Header Analysis

The email headers were analyzed using **MXToolbox Email Header Analyzer** to determine the actual sender, mail transfer path, return path, originating IP address, authentication results, and other routing information.

During the analysis, the following observations were made:

- Sender information was inconsistent.
- Return-Path differed from the displayed sender.
- The message originated from an unexpected server.
- Multiple mail hops were identified.

These indicators increased the suspicion that the email was not sent by the legitimate organization.

Headers Found	
Header Name	Header Value
Authentication-Results	spf=temperror (sender IP is 137.184.34.4) smtp.mailfrom=ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06; dkim=none (message not signed) header.d=none; dmarc=temperror action=none header.from=atendimento.com.br; compauth=fail reason=001
Received-SPF	TempError (protection.outlook.com; error in processing during lookup of ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06: DNS Timeout)
X-IncomingTopHeaderMarker	OriginalChecksum:3B61F64750F88C5569DF38A496B2374685F23D8BC662A6A19B6823B2F6745D54; UpperCasedChecksum:62071BC7A7CF5B0844A7B406B0E9EFCDDAA2CB94988E887CF8C56555AD4B52D30; SizeAsReceived:544; Count:9
Content-type	text/html; charset=UTF-8
Content-Transfer-Encoding	base64
Subject	CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!
From	BANCO DO BRADESCO LIVELO<banco.bradesco@atendimento.com.br>
To	phishing@pot
Message-Id	<20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06>
Date	Tue, 19 Sep 2023 18:35:49 +0000 (UTC)
X-IncomingHeaderCount	9
Return-Path	root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
X-MS-Exchange-Organization-ExpirationStartTime	19 Sep 2023 18:36:44.2236 (UTC)

Step 3 – SPF, DKIM and DMARC Verification

The authentication mechanisms were examined to verify the legitimacy of the sender.

- SPF Authentication Failed / Temporary Error
- DKIM Signature Missing
- DMARC Validation Failed

- Authentication Compliance Failed

The absence of valid email authentication significantly increased the probability that the message was spoofed or malicious.

Header Analyzed
Email Subject: CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!

Copy/Paste Warning
Copy/Pasting a header works for most people, but sometimes it can cause problems with things like DKIM Validation. For the best results, use our [Email Deliverability tool](#)

Delivery Information

- ✖ DMARC Compliant (No DMARC Record Found)
- ✖ SPF Alignment
- ✖ SPF Authenticated
- ✖ DKIM Alignment
- ✖ DKIM Authenticated

Relay Information
Received 57 seconds

Step 4 – Source IP Extraction

The originating IP address was extracted from the email headers.

Extracted IP Address

137.184.34.4

This IP was identified from the earliest "Received" header, representing the original sending server before the email passed through Microsoft's mail infrastructure.

There is no suspicious OLE file in attachments. N/A

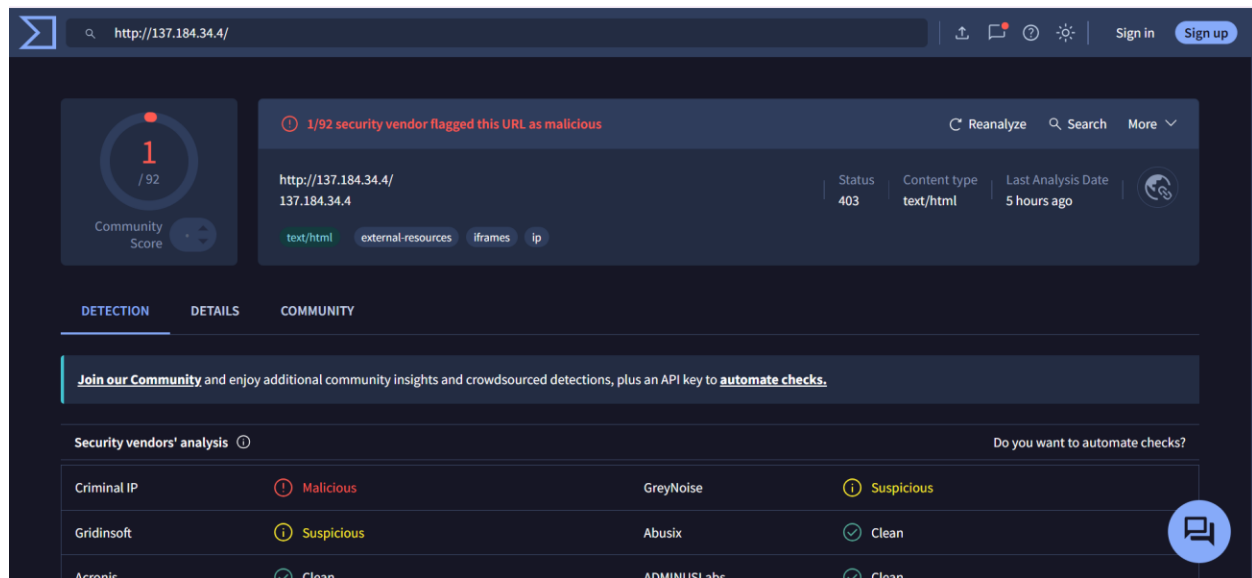
Headers

Hops

Hop	From	By	With	Date (UTC)	Delay
1				2023-09-19T18:35:49Z	N/A
2	137.184.34.4	10.13.177.138, bn8nam11ft066.mail.protection.outlook.com	microsoft smtp server (version=tls1_2, cipher=tls_ecdhe_rsa_with_aes_256_gcm_sha384) id 15.20.6813.19 via frontend transport	2023-09-19T18:36:44Z	a minute
3	bn8nam11ft066.eop-nam11.prod.protection.outlook.com, 2603:10b6:408:e6:cafe::23	2603:10b6:408:e6::28, bn0pr03ca0023.outlook.office365.com	microsoft smtp server (version=tls1_2, cipher=tls_ecdhe_rsa_with_aes_256_gcm_sha384) id 15.20.6792.28 via frontend transport	2023-09-19T18:36:45Z	a few seconds
4	2603:10b6:408:e6::28, bn0pr03ca0023.namprd03.prod.outlook.com	sa3pr19mb7370.namprd19.prod.outlook.com, 2603:10b6:806:317::17	microsoft smtp server (version=tls1_2, cipher=tls_ecdhe_rsa_with_aes_256_gcm_sha384) id 15.20.6792.27	2023-09-19T18:36:45Z	N/A

Step 5 – IP Reputation Analysis

The extracted IP address was investigated using **VirusTotal**. The reputation analysis indicated that one security vendor classified the IP as malicious, while another identified it as suspicious. Although not every vendor flagged the IP, these detections indicate that the infrastructure has been associated with suspicious activity. This finding strengthened the overall phishing assessment.



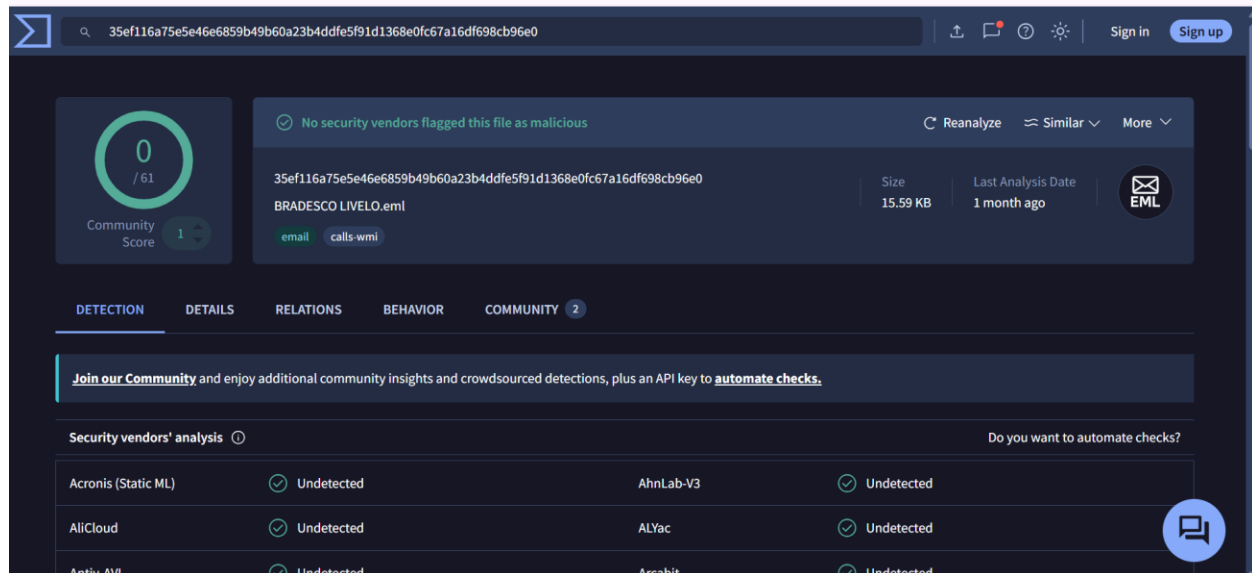
Step 6 – IOC Identification

The following Indicators of Compromise (IOCs) were collected during the investigation:

- **Sender Email Address**
⇒ `banco.bradesco@atendimento.com.br`
- **Return-Path**
⇒ `root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06`
- **Source IP Address**
⇒ `137.184.34.4`
- **Suspicious Domain**
⇒ `blog1seguimentmydomaine2bra.me`

Step 7 – Risk Assessment

Based on the overall investigation, the email was classified as a High-Risk Phishing Email. Although the email file itself was not detected as malicious by VirusTotal, multiple indicators including failed SPF, DKIM, and DMARC authentication, a suspicious source IP address, an external phishing URL, and social engineering techniques strongly suggest that the email is a phishing attempt.



Conclusion:

The investigation confirmed multiple phishing characteristics through email header analysis, authentication validation, IP reputation analysis, and URL inspection. Although the email file itself was not detected as malicious by antivirus engines, the supporting evidence including failed authentication mechanisms, suspicious infrastructure, and embedded phishing URL indicates that the email represents a phishing attempt rather than a legitimate business communication. This project demonstrates the standard workflow followed by a SOC Analyst during email triage and phishing investigations.